

Recall the definitions of π, λ -system:

- P is a π -system if: $\forall A \in P \forall B \in P A \cap B \in P$
- D is a λ -system if: non-empty, closed under complement, closed under countable disjoint union.

Teorema 1. *Let P be a π -system and let D be a λ -system, with $P \subseteq D$. Then $\sigma(P) \subseteq D$.*

Proof. Let $L := \bigcap_{D' \lambda\text{-system}, P \subseteq D'} D'$. It is immediate that $P \subseteq L$ and L is a λ -system. Fix $B \in P$ and consider $D_B := \{A \in D : A \cap B \in L\}$. Since $B \in P \subseteq L$ and P is π -system we have that $P \subseteq D_B$. Next we verify that D_B is a λ -system:

- $\Omega \cap B = B \in P \subseteq L$, that is: $\Omega \in D_B$.
- Let $A \in D_B$. Then $A \cap B \in L$ and as a consequence also $B \setminus (A \cap B) \in L$. Equivalently, $A^c \cap B \in L$ and therefore $A^c \in D_B$.
- Let $\{A_n : n < \omega\}$ be a set of disjoint elements in D_B . Then $\forall n < \omega A_n \cap B \in L$. Since $(\bigsqcup_{n < \omega} A_n) \cap B = \bigsqcup (A_n \cap B) \in L$ we get $\bigsqcup_{n < \omega} A_n \in D_B$.

Then we have $L \subseteq D_B$. Since B is an arbitrary element of P , we can say $\forall B \in P \forall A \in L A \cap B \in L$. Now fix $A \in L$ and consider $D_A := \{C \in D : C \cap A \in L\}$. From the observation above we already have $P \subseteq D_A$. Next we verify that D_A is a λ -system:

- $\Omega \cap A = A \in L$.
- $C \in D_A \Rightarrow C \cap A \in L \Rightarrow C^c \cap A \in L$ as before. Then $C^c \in D_A$.
- same as before

Again we deduce that $L \subseteq D_A$. Since A is an arbitrary element of L we get that $\forall A \in L \forall C \in L A \cap C \in L$. This means that L is a π -system. Since L also is a λ -system, we get that L is both π and λ system and therefore a σ -algebra. In particular, since $P \subseteq L \subseteq D$ we get $P \subseteq \sigma(P) \subseteq L \subseteq D$. $\square$

Teorema 2. *Let $(X_t)_{t \in T}$ stochastic process. Then TFAE:*

1. *ind incr*
2. *for all $0 \leq s \leq t$ we have $X_t - X_s$ independent from $\sigma(X_u : u \leq s)$*

Proof. • $1 \Rightarrow 2$: Consider $P := \{X_{u_1}^{-1}(A_1) \cap \dots \cap X_{u_n}^{-1}(A_n) : n < \omega, A_{u_1}, \dots, A_{u_n} \in B(\mathbb{R}^d), u_1, \dots, u_n \leq s\}$. Then P is a π -system and $\sigma(P) = \sigma(X_u : u \leq s)$. Now consider $D := \{A : \mathbb{P}(A, X_t - X_s \in B) = \mathbb{P}(A) \cdot \mathbb{P}(X_t - X_s \in B)\}$. It is immediate that D is a λ -system. Next we verify that $P \subseteq D$.

Consider $A := (X_{u_1} \in A_1, \dots, X_{u_n} \in A_n)$, arbitrary element of P and consider $f : \mathbb{R}^d \rightarrow \mathbb{R}^d, (x_1, \dots, x_n) \mapsto (x_1, x_2 - x_1, \dots, x_n - x_{n-1})$.

$$\begin{aligned}
\mathbb{P}(A, X_t - X_s \in B) &= \mathbb{P}((X_{u_1}, \dots, X_{u_n}) \in A_1 \times \dots \times A_n, X_t - X_s \in B) \\
&= \mathbb{P}(f(X_{u_1}, \dots, X_{u_n}) \in f[A_1 \times \dots \times A_n], X_t - X_s \in B) \\
&= \mathbb{P}((X_{u_1}, \dots, X_{u_n} - X_{u_{n-1}}) \in f[A_1 \times \dots \times A_n], X_t - X_s \in B) \\
&= \mathbb{P}((X_{u_1}, \dots, X_{u_n} - X_{u_{n-1}}) \in f[A_1 \times \dots \times A_n]) \cdot \mathbb{P}(X_t - X_s \in B) \\
&= \mathbb{P}(f(X_{u_1}, \dots, X_{u_n}) \in f[A_1 \times \dots \times A_n]) \cdot \mathbb{P}(X_t - X_s \in B) \\
&= \mathbb{P}((X_{u_1}, \dots, X_{u_n}) \in A_1 \times \dots \times A_n) \cdot \mathbb{P}(X_t - X_s \in B)
\end{aligned}$$

- $2 \Rightarrow 1$: maybe later

□